



MAYSSA BEN ROMDHANE

Full-Stack Software Engineer

Gabès, Tunisia | +216 58 936 265 | benromdhanemayssa23@gmail.com

GitHub: MayssaBr24 | LinkedIn: mayssa-ben-romdhane

PROFESSIONAL SUMMARY

Software Engineer specializing in **microservices architectures** and Full-Stack development (**Spring Boot, MERN, Angular, .NET, Laravel**). Expert in intelligent automation via **n8n** and the **MCP protocol**. Proficient in **Docker containerization**, Big Data (**Hadoop, Spark, Kafka**), and **IoT** solutions for secure production environments.

TECHNICAL SKILLS

Frameworks & Dev: Spring Boot, MERN Stack, Angular, Laravel, .NET 8, Vue.js, Blazor, Flutter, FastAPI.
Architecture: Microservices, Clean Architecture (4 layers), Event-Driven, RESTful APIs, Eureka, Swagger.
AI & Automation: n8n (Workflow Automation), MCP (Model Context Protocol), Autonomous AI Agents, LLM Integration.
DevOps & Cloud: Docker (Containerization), Docker Compose, Kubernetes (K8s), Minikube, Azure, Nginx.
Middleware & Data: RabbitMQ (Message Broker), Kafka, Hadoop, Spark, Redis, JWT, mTLS, SSL/TLS.
Databases: PostgreSQL, MongoDB, SQL Server, MySQL, Firebase, Entity Framework, Hibernate.
IoT & Languages: Python, Java, C#, PHP, Raspberry Pi, Selenium (QA Automation), Postman.

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer - HealthFit Project

June 2025 – September 2025

Technologies: Laravel, Vue.js, Tailwind CSS, Azure App Services, Docker

- Designed a membership management and class scheduling platform with **3 specialized dashboards**.
- Developed modern, reactive interfaces using **Vue.js** and **Tailwind CSS**.
- Managed application architecture via **Docker containerization** to ensure service isolation.
- Deployed and configured container instances directly on **Azure App Services**.

Full-Stack Web Developer - ESSAT Finance Project

January 2025 – June 2025

Technologies: Spring Boot, React, mTLS, Nginx, Microservices

- Designed a transaction management module within a distributed **microservices architecture**.
- Implemented inter-service communication and financial data synchronization.
- Applied **mTLS authentication** to the Finance microservice to restrict access to a single authorized client.
- Optimized financial reporting via asynchronous queries to ensure high module availability.

Network Infrastructure Intern

August 2024

Organization: Tunisie Télécom - Network Maintenance Service

- **Diagnosed and resolved** complex WiFi connectivity issues for residential customers.
- Conducted on-site interventions for technical **troubleshooting** and hardware configuration of client modems.
- Analyzed network performance and **optimized Internet connections** using monitoring tools.

TECHNICAL PROJECTS

Artificial Intelligence & Advanced Automation

2025

Technologies: n8n, MCP Protocol, MERN Stack, AI Agents

- **n8n Workflow:** Created complex automation flows linking business services and productivity tools.
- **MERN Ecosystem:** Developed a control and monitoring interface for AI agents using React and Node.js.
- **MCP Protocol:** Implemented the Model Context Protocol to interface intelligent agents with SQL/NoSQL databases securely and contextually.

Microservices Architecture & Messaging - .NET 8

2025

Technologies: ASP.NET Core, Blazor, Vue.js, RabbitMQ, JWT, Clean Architecture, Swagger

- Designed a distributed system based on **Clean Architecture** principles with a decoupled frontend using **Blazor** and **Vue.js**.
- Implemented **RabbitMQ** message broker to ensure asynchronous and decoupled communication between microservices.
- Secured endpoints via **JWT** and created interactive API documentation with **Swagger**.
- Developed a unified management interface to monitor microservices and message flows in real-time.

IoT & Smart Embedded Systems

2025

Technologies: Python, Raspberry Pi, Sensors

- Developed an autonomous vehicle featuring **obstacle detection** and **line follower** capabilities.
- Integrated precision sensors for real-time monitoring: **humidity**, **temperature**, and **distance**.

Big Data & Massive Stream Processing

2025

Technologies: Hadoop, Apache Spark, Kafka, Distributed Systems

- Established a **Hadoop** ecosystem for distributed storage and processing of large data volumes.
- Developed a real-time streaming pipeline using **Kafka** for ingestion and **Spark** for analytics.

EDUCATION & LEADERSHIP

Engineering Degree in Software Engineering | ESSAT Gabès

2022 – Present

Certifications: MERN Stack, Big Data & Machine Learning (IBM), ASP.NET.

Leadership: Communications Manager at Club Ideal | Active IEEE Member.

LANGUAGES

Arabic (Native), **French** (C2 - Bilingual), **English** (B2 - Upper-Intermediate).